AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TT284
Web Technologies

Final Examination
Fall Semester 2025/2026

Date: 15/Jan./2026

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 3 hours

Instructions:

- Total Marks: 100
- This exam consists of 4 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART 1

(16 marks - 8 True/False questions - 2 marks each question)

1. Pixel size refers to the physical dimensions of the screen typically measured on the diagonal.

2. A phone with a stand-alone GPS module allows it to work with the phone network to give a quicker fix than a built-in GPS, and a more accurate one than the phone network or Wifi triangulation alone.

3. Once an app is developed, the next stage is to make it available either within an organization or to the general public. If they are for the general public, a choice has to be made as to whether to charge or not. The process of making them available to the market is called: the 'route to market'.

4. If a system has not yet been built, the normal calculation for availability cannot be applied, so we have to turn to a slightly different notion of availability based on the mean time to failure (MTTF) and the 'mean time to repair' (MTTR), also known as 'mean time to replace'.

5. Risk management is a field of study that brings together a wide range of techniques and tools for managing any development activity. It can help to organise the tasks and manage the resources so that an application to organise the tasks and manage the resources so that an application gets completed on time, on budget, and most importantly, fulfils its requirements.

6. System tests, or end-to-end tests, are intended to verify the interoperability and performance of the application in its final configuration and to verify integration with other hardware and software systems that support the wider business functions.

7. No matter how good the installation and configuration process, errors are made and applications are still subject to attacks that attempt to exploit new vulnerabilities. It is therefore essential to record all the activities that take place on a server so that we can assess if anyone is attempting to attack our application.

8. Symmetric encryption methods use a pair of keys; one called the public key, and the other the private key. The two keys are mathematically related in a special way such that anything encrypted with the private key can be decrypted with the public key and vice versa. However, neither key can decrypt anything that it previously encrypted.

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PART 2

(9 marks - 3 Short essay questions - 3 marks each question)

Question 1: (3 marks)
List three ways that show how can the mobile device 'know' where it is?

Question 2: (3 marks)
List three of the potential benefits of virtualisation.

Question 3: (3 marks)
List three of the top security risks for web applications.

PART 3

(25 marks - 5 Long essay questions - 5 marks each question)

Question 1: (5 marks)
IT projects have gained a reputation for grand failures, studies have shown some common features. Discuss these common features.

Question 2: (5 marks)
One of the challenges of the user requirements phase is satisfying conflicting features or functions. Individual stakeholders will have different perspectives on what the project should achieve and hence what the priorities should be. Discuss the levels of priorities on what the project should achieve.

Question 3: (5 marks)
Discuss the methods that can be used to derive the resource and duration estimates of a project.

Question 4: (5 marks)
Discuss the Stress test of the load performance testing.

Question 5: (5 marks)
Discuss two traditional methods of authentication.

PART 4

(50 marks - 5 Problem Solving Questions - 10 marks each question)

Question 1: (10 marks)
Disk Drive manufacturers test a sample of 100 drives by running them for a period of 2000 hours. At the end of that period two drives were found to have failed. Calculate the 'mean time to failure' (MTTF).

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Question 2: (10 marks)
What is the output of the following program?

```
<html>  <body>  <script>
function Follow(x , y) {
for (counter = 10 ; counter>=1 ; counter = counter -2) {
  if (y >= -10 && x <=15)  {
  document.write(x + "<br>");
  x = x + 2;
  y = y - 2;  }
else  {
  document.write(y + "<br>");
  y = y + 3;
  x = x + 3  } } }
Follow(10,-2);
</script>  </body>  </html>
```

Question 3: (10 marks)
Write HTML code that will create the table shown in Figure 1.

Students Grades

Name	Score	Grade
Alice	92	A
Belal	85	B
Dalal	78	C
George	65	D

Figure 1

Question 4: (10 marks)
Write a JavaScript program that classifies forum members based on their Number of participations, Number of years as a member, Number of admin notices, and Compliance and Ethics score as follows:

1. Golden Member:
   1.1. Number of participations >= 300
   1.2. Number of years as a member > 3
   1.3. Number of admin notices <=1
2. Silver Member:
   2.1. Number of participations >= 200
   2.2. Number of years as a member > 2
   2.3. Number of admin notices <= 2.
3. Bronze Member:
   3.1. Number of participations >= 100
   3.2. Number of years as a member > 1
   3.3. Number of admin notices <= 3

Note:
All members must have a minimum Compliance and Ethics score of 4 out of 5 to qualify for any level.

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Question 5: (10 Marks)

Develop a simple web page containing the following specifications:

1. Page Title "Teaching Feedback Page"
2. Header 1 "Teaching Feedback Form"
3. Form action to locate the script that will process the form data "http//norton.open.ac.uk/reflect.php"
4. Image that displays the logo that was saved in the Logo.GIF file.
5. Text box for Tutor First Name.
6. Text box for Tutor Last Name
7. Text box for Training Course.
8. Fieldset1 (Usefulness) Radio button Excellent.
9. Fieldset1 (Usefulness) Radio button Average.
10. Fieldset1 (Usefulness) Radio button Poor.
11. Fieldset2 (Teaching) Radio button Excellent.
12. Fieldset2 (Teaching) Radio button Average.
13. Fieldset2 (Teaching) Radio button Poor.
14. Fieldset3 Checkbox for the Tutor.
15. Fieldset3 Checkbox for the Content.
16. Fieldset3 Checkbox for the Training.
17. Text area for the description of special requests.
18. Link to go to "Training Institute" web page "http://www.Training.com".
19. Submit button with Submit value.
20. Reset button with Reset value.

The web page should look like the following design (Figure 2):

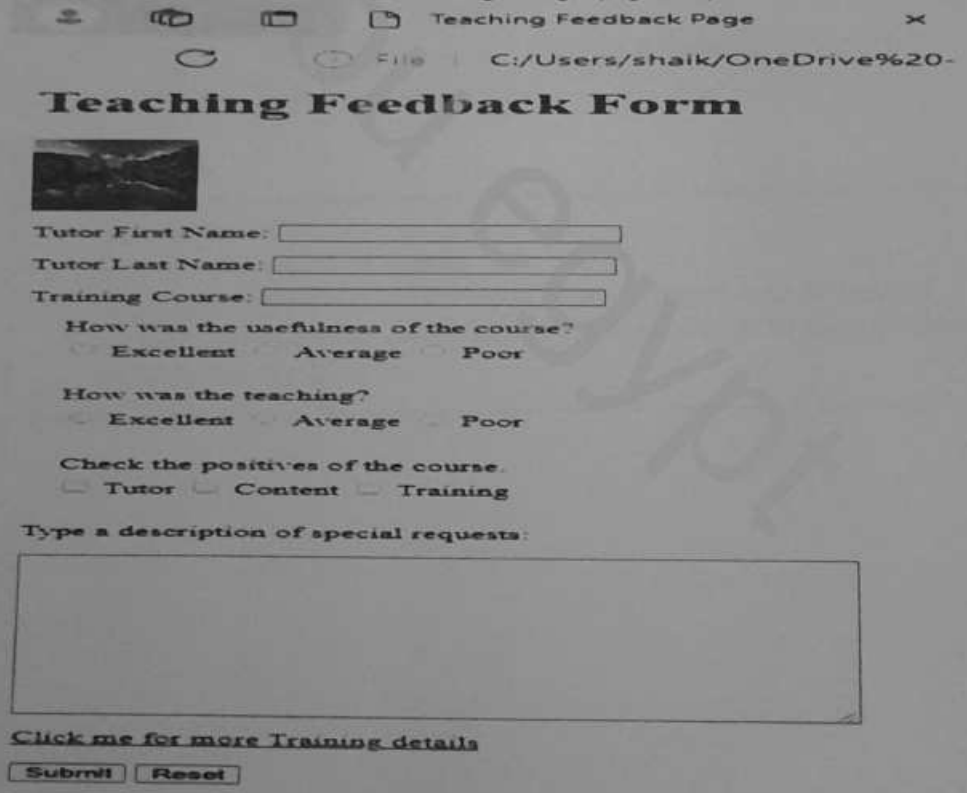


Figure 2

End of questions

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